



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

November 05, 2021

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 11382
(THIRTEENTH REVISION)

EXPIRATION DATE: 2025-09-30

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Structural Composites Industries Llc
Pomona, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of non-DOT specification fiber reinforced plastic (FRP) hoop wrapped cylinders to be used for the transportation in commerce of the materials listed in paragraph 6 below. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.209, §§ 173.302(a)(1) and 173.304(a) in that non-DOT specification cylinders are not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Structural Composites Industries LLC dated August 21, 2017, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/Divison	Identification Number	Packing Group
Air, compressed (containing up to 39% by volume oxygen)	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Carbon dioxide	2.2	UN1013	N/A
Helium, compressed	2.2	UN1046	N/A
Methane, compressed <i>or</i> Natural gas, compressed (<i>with high Methane content</i>)	2.1	UN1971	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Oxygen, compressed	2.2	UN1072	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING - Prescribed packaging is a non-DOT specification fiber reinforced plastic (FRP) hoop wrapped (HW) aluminum cylinder made in conformance with the following:

(1) Structural Composites Industries' drawing number 1272582, dated December 16, 1994; the information provided in the Report of Design Qualification Testing, dated October 26, 1994, on file with the Office of Hazardous Materials Safety Approvals and Permits Division (OHMSAPD); and

(2) DOT FRP-2 Standard (§ 178.BB), Revision 1 dated January 4, 1987, except as follows:

§ 178.BB-1 Compliance.

Cylinders authorized under this special permit must be in compliance with the basic and specific requirements prescribed in this special permit.

Design qualification test results must be on file with the OHMSAPD at least 30 days prior to shipment of the first lot for each cylinder design meeting the definition of design change in § 178.BB-18.

§ 178.BB-2 Type, size and service pressure.

Type 3HW composite cylinder consisting of resin impregnated continuous filament windings in circumferential or near circumferential directions over a seamless aluminum liner made in compliance with § 178.BB-6(a), not over 100 pounds water capacity, and a service pressure not less than 900 psi and not greater than 2220 psi.

§ 178.BB-4 Duties of the Inspector.

(a) Verify conformance with the requirements of this special permit and with DOT FRP-2 Standard dated January 15, 1982.

(b) Verify conformance of aluminum liner with § 178.BB-6. Verify conformance of the filament and resin system components with the requirements specified in § 178.BB-5 of this special permit.

(c) thru (g) * * *

§ 178.BB-5 Authorized Materials and Identification of materials.

(a) Aluminum liner must be 6061 alloy of T6 temper.

(b) thru (e) * * *

§ 178.BB-6 Manufacture:

(a) * * *

(1) and (2) * * *

(3) Starting material is sheet, or plate traceable to cast stock by cast code number, and manufacturer, or seamless tubing formed by the die and mandrel method from cut billets as described in SCT's letter dated May 25, 1983 (enclosure 3) on file with the OHMSAPD.

(4) Ultrasonic inspection of sheet or plate stock is not required. Inspection of cast hollow ingot must be as described in the application (enclosure 3).

(5) Cylinder shells must be manufactured by deep drawing or by drawing and ironing the sheet or plate without external heating of the material or by extrusion of cut billets in accordance with the methods described in the application. Spun closures are not authorized. Cylinders may be designed with plugged spun base. Plugging to correct manufacturing defects is not permitted. End contour must be hemispherical or ellipsoidal with ratio of major to minor axis not exceeding two to one with concave side to pressure.

(b) thru (e) * * *

§ 178.BB-12 Destructive tests:

(a) Cycling tests:

(1) One cylinder taken at random out of each lot of 200 cylinders must be subjected to cycling pressurization test by hydrostatically pressurizing the cylinder between approximately zero psig and the designated pressure at a rate not to exceed 10 cycles per minute. Adequate recording instrumentation must be provided if equipment is to be left unattended for period of time. All cylinders used in the cycling test must be destroyed.

(c) * * *

§ 178.BB-18 Design qualification tests.

(a) thru (c) * * *

(d) Pressure cycling test. All pressure cycling tests will be performed by hydrostatically pressurizing the test cylinder between approximately zero psig and the designated pressure at a rate not exceeding 10 cycles per minute. All cylinders used in cycle tests must be destroyed. Adequate recording instrumentation must be provided if equipment is to be left unattended for periods of time.

(d)(1) through (d)(3) * * *

(e) through (h) * * *

b. SAFETY CONTROL MEASURES -

(1) Cylinders may not be used for underwater breathing purposes.

- (2) Cylinders used in oxygen service must conform with § 173.302a(b)(5)(1) through (b)(5)(4).
- (3) Cylinder service life may not exceed 15 years from the date of manufacture as marked on the cylinder.
- (4) A cylinder that has been subjected to fire may not be returned to service.
- (5) Transportation of methane by cargo-only aircraft is limited to 150 kilograms (330 pounds net quantity limitation per cylinder).
- (6) Cylinders must be packaged in accordance with § 173.301(a)(9).
- (7) The cylinders described in this special permit are authorized only for normal transportation as an article of commerce (i.e., the movement of hazardous materials packages from consignor to consignee). No cylinder may be transported in a partially filled condition other than full or empty per § 173.29, unless it is equipped with a pressure relief device designed, manufactured and tested for partially filled cylinders.
- (8) Filling requirements are subject to all terms contained in § 173.302a for 3AL specification cylinders.

c. MARKING - Compliance with § 178.BB-15 is required.

d. TESTING - Each cylinder must be reinspected and hydrostatically retested every three years in accordance with § 180.213(a)(2), as prescribed for DOT 3HT specification cylinders, except that the rejection elastic expansion does not apply and the permanent volumetric expansion may not exceed 5 percent of the total volumetric expansion at test pressure. Retest dates must be stamped on the exposed metallic surface of the cylinder neck or marked on a label securely affixed to the cylinder and overcoated with epoxy near the original test date. Reheat treatment or repair of rejected cylinders is not authorized. When a hydrostatic retest is repeated as provided for in § 180.205(f), only two such retests are permitted.

8. SPECIAL PROVISIONS:

a. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.

- b. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.
 - c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
 - d. Each packaging manufactured under the authority of this special permit must be marked with a registration symbol designated by the Office of Hazardous Materials Special Permits and Approvals for a specific manufacturing facility.
 - e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit. It must be made available to a DOT representative upon request.
 - f. Transportation of flammable gases is not authorized aboard passenger-carrying aircraft or cargo vessel.
 - g. Transportation of oxygen is only authorized when in accordance with § 175.501.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, cargo-only aircraft and passenger-carrying aircraft. (See paragraphs 7.b.(5), 8.f. and 8.g. of this special permit.)
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel and aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each "Hazmat employee", as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) - 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term "exemption" to "special permit" and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 Immediate notice of certain hazardous materials incidents, and 171.16 Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: Kenny Herzog